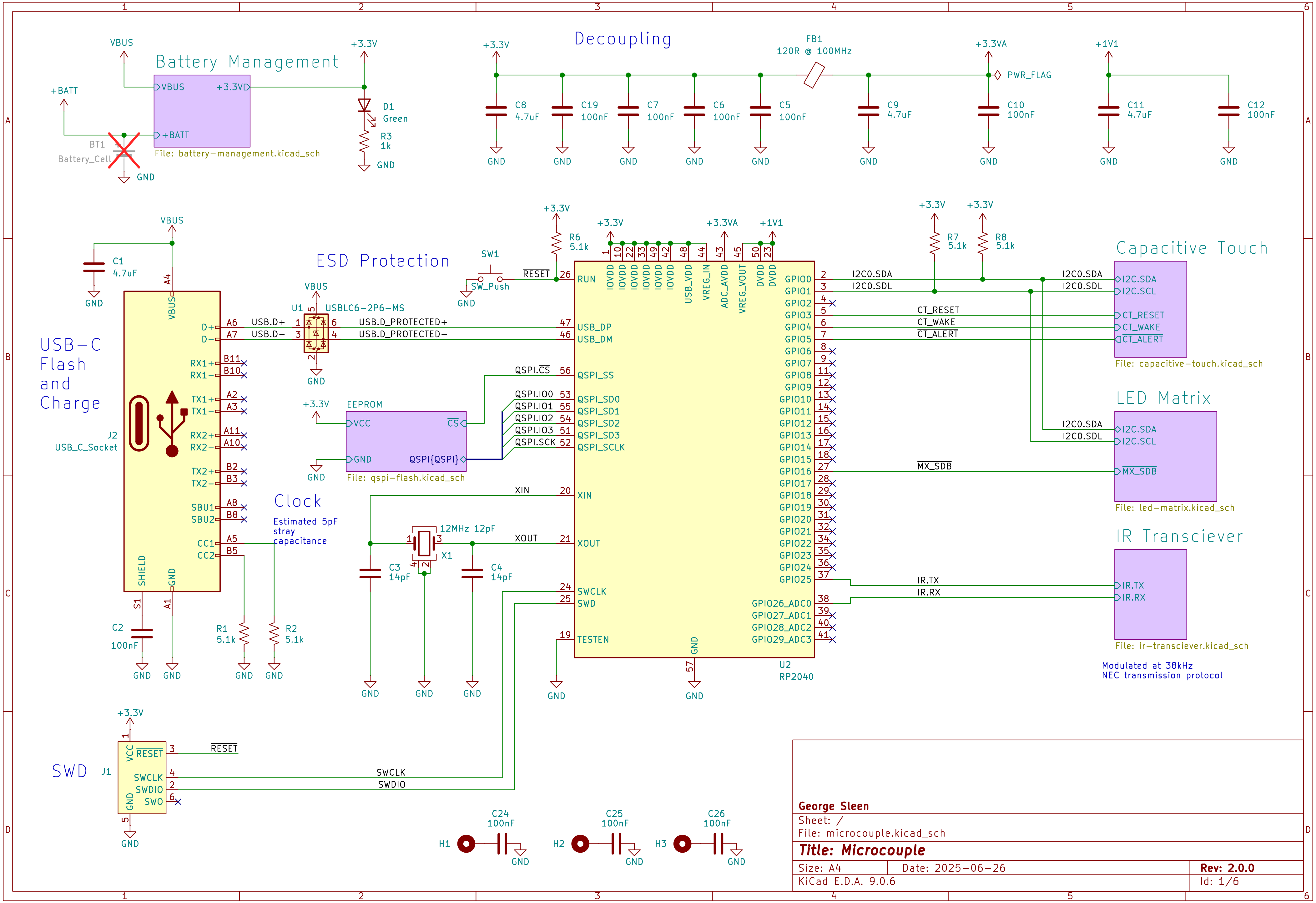
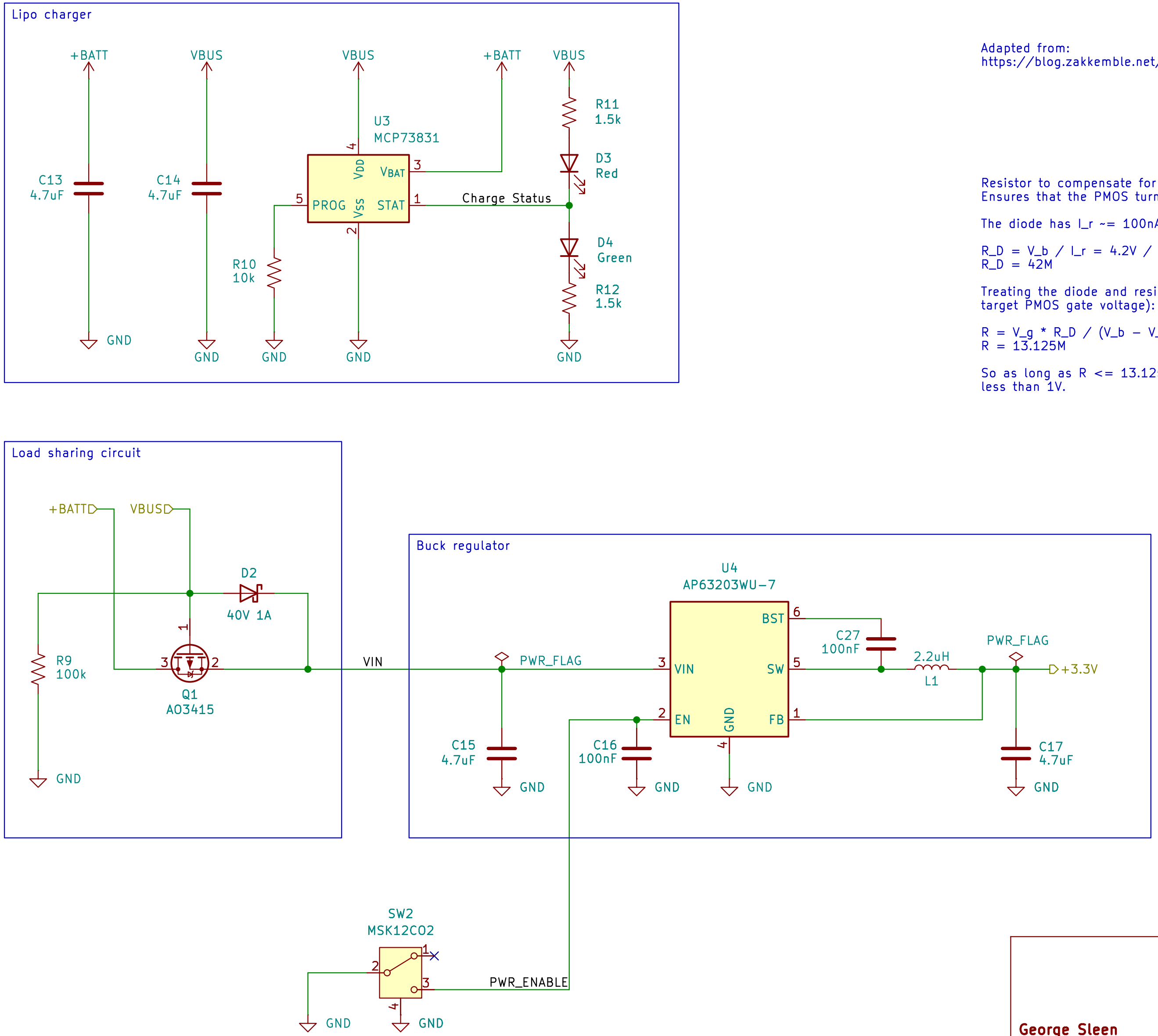


Battery Management
VBUS
+3.3V
+BATT
BT1
Battery_Cell
GND
File: battery-management.kicad_sch
D1
Green
R3
1k
Decoupling
FB1
120R @ 100MHz
+3.3VA
PWR_FLAG
+1V1
C8 4.7uF
C19 100nF
C7 100nF
C6 100nF
C5 100nF
C9 4.7uF
C10 100nF
C11 4.7uF
C12 100nF
USB-C Flash and Charge
J2
USB_C_Socket
C1 4.7uF
C2 100nF
R1 5.1k
R2 5.1k
ESD Protection
U1 USBLC6-2P6-MS
USB.D+
USB.D-
USB.D_PROTECTED+
USB.D_PROTECTED-
EEPROM
File: qspi-flash.kicad_sch
QSPI{QSPI}
Clock
Estimated 5pF stray capacitance
12MHz 12pF
X1
C3 14pF
C4 14pF
SW1
SW_Push
R6 5.1k
RESET
U2
RP2040
R7 5.1k
R8 5.1k
I2C0.SDA
I2C0.SDL
CT_RESET
CT_WAKE
CT_ALERT
MX_SDB
IR.TX
IR.RX
Capacitive Touch
File: capacitive-touch.kicad_sch
LED Matrix
File: led-matrix.kicad_sch
IR Transciever
File: ir-transciever.kicad_sch
Modulated at 38kHz
NEC transmission protocol
SWD
J1
SWCLK
SWDIO
SWO
H1 C24 100nF
H2 C25 100nF
H3 C26 100nF
George Sleen
Sheet: /
File: microcouple.kicad_sch
Title: Microcouple
Size: A4
Date: 2025-06-26
KiCad E.D.A. 9.0.6
Rev: 2.0.0
Id: 1/6



Adapted from:
https://blog.zakkemble.net/a-lithium-battery-charger-with-load-sharing/

Resistor to compensate for reverse leakage current in the diode.
Ensures that the PMOS turns back on properly.

The diode has I_r ~= 100nA @ 5V

R_D = V_b / I_r = 4.2V / 100nA
R_D = 42M

Treating the diode and resistor like a voltage divider (with a 1V target PMOS gate voltage):

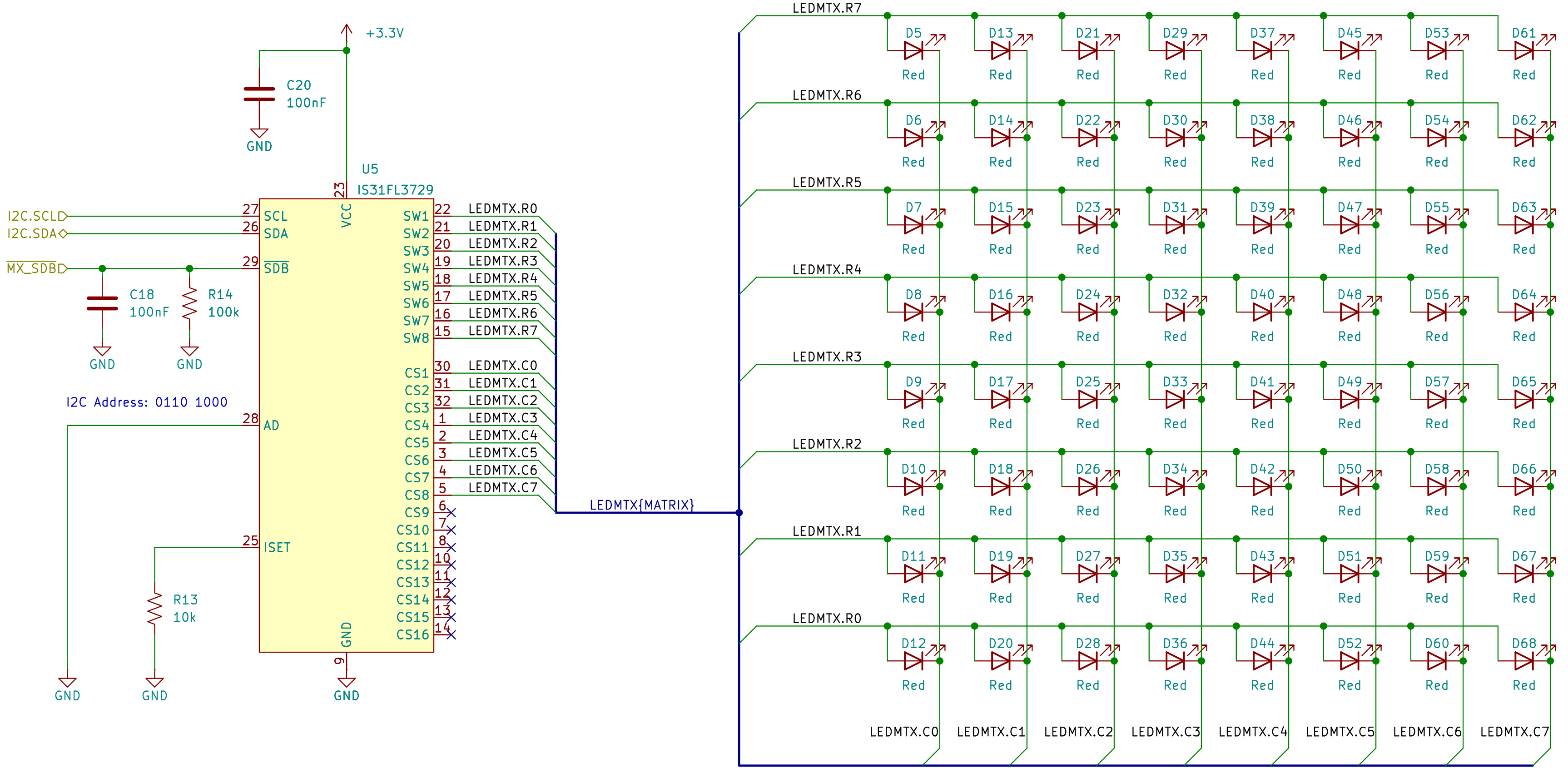
R = V_g * R_D / (V_b - V_g) = 1V * 42M / (4.2V - 1V)
R = 13.125M

So as long as R <= 13.125M the gate voltage will be kept at less than 1V.

George Sleen
Sheet: /Battery Management/
File: battery-management.kicad_sch

Title: Battery Management

Size: A4	Date: 2025-04-27	**Rev: 2.0.0**
KiCad E.D.A. 9.0.6		Id: 2/6



George Sleen

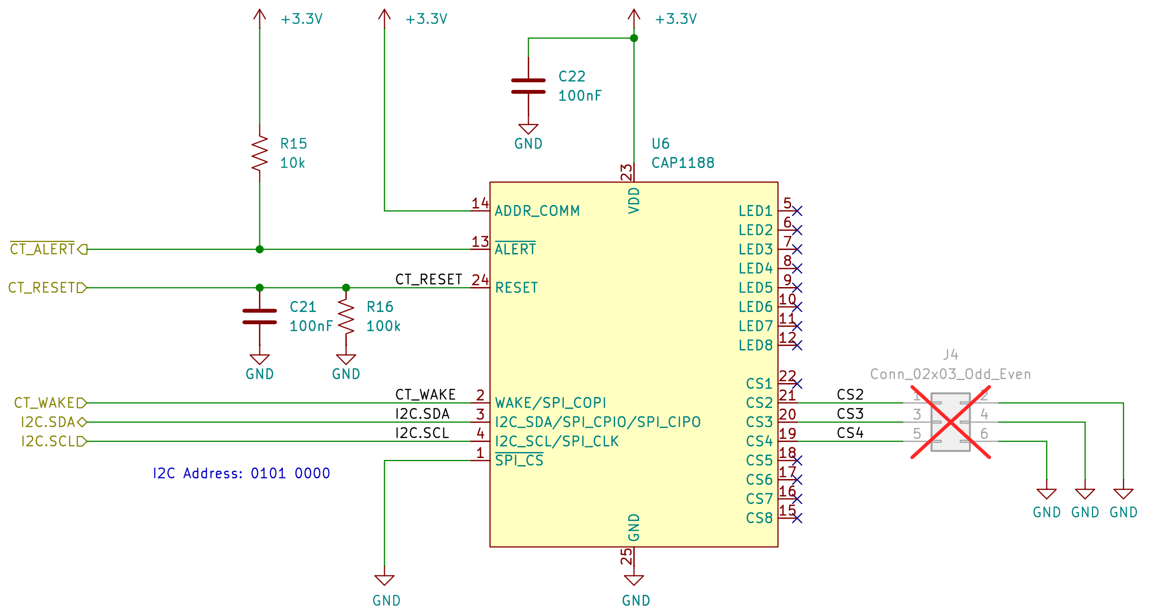
Sheet: /LED Matrix/

File: led-matrix.kicad_sch

Title: LED Matrix

Size: A4	Date: 2025-04-27	**Rev: 2.0.0**
KiCad E.D.A. 9.0.6		Id: 3/6

Sensor design based on:
https://www.slideshare.net/slideshow/capacitance-sensing-layout-guidelines-for-psoc-capsense/1007241



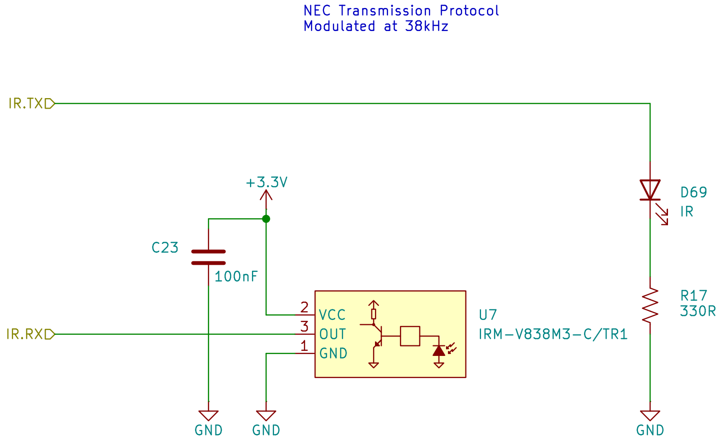
George Sleen

Sheet: /Capacitive Touch/

File: capacitive-touch.kicad_sch

Title: Capacitive Touch

Size: A4	Date: 2025-04-27	**Rev: 2.0.0**
KiCad E.D.A. 9.0.6		Id: 4/6

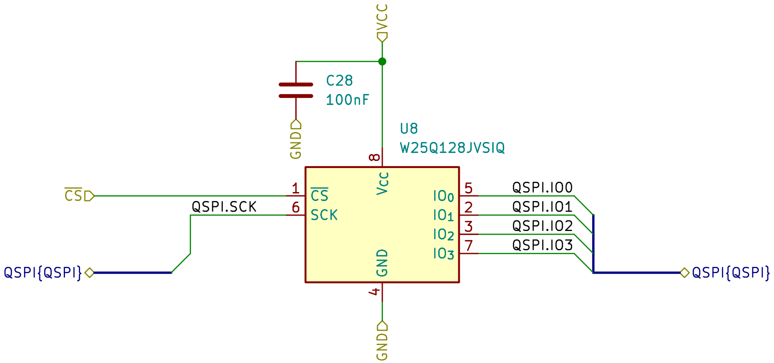


George Sleen

Sheet: /IR Transciever/
File: ir-transciever.kicad_sch

Title: IR Transciever

Size: A5	Date: 2025-04-27	**Rev: 2.0.0**
KiCad E.D.A. 9.0.6		Id: 5/6



George Sleen

Sheet: /EEPROM/

File: qspi-flash.kicad_sch

Title: 128Mbit Flash

Size: A5	Date: 2025-06-26	**Rev: 2.0.0**
KiCad E.D.A. 9.0.6		Id: 6/6